Tue
Example

$y' = Ay + b$ $\qquad A = \begin{bmatrix} 1 & 4 \\ 3 & 7 \end{bmatrix}$

$b = \begin{bmatrix} 7 \\ 9 \end{bmatrix}$

Guess $\quad y_p = u$

$y_p' = 0 \qquad 0 = Au + \begin{bmatrix} 7 \\ 9 \end{bmatrix}$

Lin Sys to solve $\quad Au = \begin{bmatrix} -7 \\ -9 \end{bmatrix}$

$y' = Ay + \begin{bmatrix} 6 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} t$

Guess $\quad \vec{y}_p = \vec{u}_0 + \vec{u}_1 t \qquad 0 + \vec{u}_1 = Au_0 + tAu_1 + \begin{bmatrix} 6 \\ 2 \end{bmatrix} + t\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$\vec{y}_p' = 0 + \vec{u}_1$

$Au_1 + \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\vec{u}_1 \cancel{A} = Au_0 + \begin{bmatrix} 6 \\ 2 \end{bmatrix}$